Pearson Edexcel

Mark Scheme

Mock Set 3

Pearson Edexcel GCE Mathematics

Advanced Subsidiary Level in Mathematics

Paper 01 8MA0/01 Pure Mathematics

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

EDEXCEL GCE MATHEMATICS
General Instructions for Marking

1. The total number of marks for the paper is 100.

2. The Edexcel Mathematics mark schemes use the following types of marks:
   - **M** marks: method marks are awarded for 'knowing a method and attempting to apply it', unless otherwise indicated.
   - **A** marks: Accuracy marks can only be awarded if the relevant method (M) marks have been earned.
   - **B** marks are unconditional accuracy marks (independent of M marks)
   - Marks should not be subdivided.

3. Abbreviations
   These are some of the traditional marking abbreviations that will appear in the mark schemes.
   - bod – benefit of doubt
   - ft – follow through
   - the symbol $\sqrt{}$ will be used for correct ft
   - cao – correct answer only
   - cso - correct solution only. There must be no errors in this part of the question to obtain this mark
   - isw – ignore subsequent working
   - awrt – answers which round to
   - SC: special case
   - oe – or equivalent (and appropriate)
   - dep – dependent
   - indep – independent
   - dp decimal places
   - sf significant figures
   - $*$ The answer is printed on the paper
   - $\sqsubset$ The second mark is dependent on gaining the first mark

4. For misreading which does not alter the character of a question or materially simplify it, deduct two from any A or B marks gained, in that part of the question affected.

5. Where a candidate has made multiple responses <u>and indicates which response they wish to submit</u>, examiners should mark this response.
   If there are several attempts at a question <u>which have not been crossed out</u>, examiners should mark the final answer which is the answer that is the <u>most complete</u>.

6. Ignore wrong working or incorrect statements following a correct answer.

7. Mark schemes will firstly show the solution judged to be the most common response expected from candidates. Where appropriate, alternatives answers are provided in the notes. If examiners are not sure if an answer is acceptable, they will check the mark scheme to see if an alternative answer is given for the method used.

Question	Scheme	Marks	AOs
1(a)	$x^3 \to x^2$ or $5x \to 5$ or $x^{-1} \to x^{-2}$	M1	1.1b
	Two of $+\frac{9}{2}x^2,\ -5,\ +\frac{10}{x^2}$	A1	1.1b
	$\frac{dy}{dx} = \frac{9}{2}x^2 - 5 + \frac{10}{x^2}$	A1	1.1b
		(3)	
(b)	$\left(\frac{dy}{dx} =\right) \frac{9}{2}(-2)^2 - 5 + \frac{10}{(-2)^2} = \ldots \left(= \frac{31}{2}\right)$	M1	1.1b
	$"\frac{31}{2}" \to "-\frac{2}{31}"$	dM1	1.2
	$y - 3 = "-\frac{2}{31}"(x+2)$	dM1	1.1b
	$2x + 31y - 89 = 0$	A1	1.1b
		(4)	
			(7 marks)

Notes

(a)

M1: Reduces the power of x by one on one of the terms (indices do not need to be processed)

A1: Two correct unsimplified terms (may be given in a list)

A1: $\frac{9}{2}x^2 - 5 + \frac{10}{x^2}$ or simplified equivalent e.g. $4.5x^2 - 5 + 10x^{-2}$

(b)

M1: Attempts to find the gradient of the curve at $x = -2$

dM1: Finds the negative reciprocal of the gradient found at $x = -2$. It is dependent on the first method mark.

dM1: Attempts to find the equation of the normal using a changed gradient at $(-2, 3)$. If they use $y = mx + c$ they must proceed as far as $c = \ldots$. It is dependent on the first method mark.

A1: $2x + 31y - 89 = 0$ or any integer multiple of this equation

Question	Scheme	Marks	AOs
2	$3x^2-7x-10>0 \Rightarrow (3x-10)(x+1)>0 \Rightarrow \text{CVs}=-1,\ \dfrac{10}{3}$	M1	1.1b
	Attempts the outside region: $x<\text{"}-1\text{"}$ and $x>\text{"}\dfrac{10}{3}\text{"}$	M1	1.1b
	e.g. $\{x: x<-1\}\cup\left\{x: x>\dfrac{10}{3}\right\}$	A1	2.5
		(3)	
			(3 marks)

Notes

M1: Attempts to find the critical values for the quadratic inequality by factorising, completing the square or quadratic formula (an algebraic method). They cannot just state the roots.

M1: Attempts the outside region for their two critical values. Condone $\leq \geq$ signs for this mark. Also condone incorrect combining of the inequalities such as $\text{"}\dfrac{10}{3}\text{"}<x<\text{"}-1\text{"}$

A1: $\{x: x<-1\}\cup\left\{x: x>\dfrac{10}{3}\right\}$ or equivalent using set notation.

Note $\{x: x<-1\}\cup\left\{x: x>\dfrac{10}{3}\right\}$ with no working scores M0M1A0

Question	Scheme	Marks	AOs
3(a)	$\mathrm{f}(2) = 6(2)^3 - (2a+5)(2)^2 + 21(2) + a = 0$	M1	1.2
	$48 - 8a - 20 + 42 + a = 0 \Rightarrow a = 10$ *	A1*	2.4
		(2)	
(b)	At least two of $p = 6,\ q = -13,\ r = -5$ for $(x-2)(px^2 + qx + r) = 0$	M1	3.1a
	$6x^2 - 13x - 5$	A1	1.1b
	$(3x+1)(2x-5) = 0 \Rightarrow x = ...$	M1	1.1b
	$x = -\frac{1}{3}, \frac{5}{2}, 2$	A1	2.2a
		(4)	
(c)	$x = \frac{1}{3}$	B1ft	2.2a
		(1)	
			(7 marks)

Notes

(a)

M1: Substitutes 2 into the expression for $\mathrm{f}(x)$ and sets equal to 0

A1*: Rearranges to achieve the given answer with at least one intermediate stage of working seen.

(b)

M1: Uses $(x-2)$ to find the quadratic factor $px^2 + qx + r$. Implied by at least two correct constants.

A1: $6x^2 - 13x - 5$

M1: Attempts to solve their quadratic = 0 by factorising, completing the square or using the formula. It cannot be just from stating the values from a calculator.

A1: $x = -\frac{1}{3}, \frac{5}{2}, 2$ or equivalent

(c)

B1ft: $x = \frac{1}{3}$ or follow through their smallest value from (b)

Question	Scheme	Marks	AOs
4(a)	$\sqrt{3^2+12^2}$	M1	1.1b
	$=3\sqrt{17}$	A1	1.1b
		(2)	
(b)	$6b-9a=-3$ and $15a+4b=12 \;\rightarrow\; a=...$ or $b=...$	M1	3.1a
	One of $a=\frac{2}{3}$ or $b=\frac{1}{2}$	A1	1.1b
	$a=\frac{2}{3}$ and $b=\frac{1}{2}$	A1	1.1b
		(3)	
			(5 marks)

Notes

(a)

M1: Attempts Pythagoras' theorem correctly by squaring and adding before taking the square root.

A1: $3\sqrt{17}$ cao

(b)

M1: Forms and solves two correct simultaneous equations leading to a value for *a* or *b*. Condone sign slips.

A1: $a=\frac{2}{3}$ or $b=\frac{1}{2}$

A1: $a=\frac{2}{3}$ and $b=\frac{1}{2}$

Question	Scheme	Marks	AOs
5(i)	$3(x-2\sqrt{5})=x\sqrt{5}\Rightarrow 3x-x\sqrt{5}=6\sqrt{5}\Rightarrow x(.......)=6\sqrt{5}$	M1	1.1b
	$x\left(3-\sqrt{5}\right)=6\sqrt{5}\Rightarrow x=\dfrac{6\sqrt{5}}{3-\sqrt{5}}\times\dfrac{3+\sqrt{5}}{3+\sqrt{5}}=...$	M1	1.1b
	$x=\dfrac{9\sqrt{5}+15}{2}$	A1	1.1b
		(3)	
(ii)	$e^{4x-1}=5e^{\frac{1}{2}x}\Rightarrow e^{\frac{7}{2}x-1}=5$ or $4x-1=\ln\left(5e^{\frac{1}{2}x}\right)$	M1	1.1b
	$\dfrac{7}{2}x-1=\ln 5\Rightarrow x=...$ or $4x-1=\ln 5+\dfrac{1}{2}x\Rightarrow x=...$	M1	1.1b
	$x=\dfrac{2}{7}\left(1+\ln 5\right)$	A1	1.1b
		(3)	
			(6 marks)

Notes

(i)

M1: Multiplies out the brackets, collects terms in x on one side and attempts to take out a factor of x

M1: Attempts to rationalise the denominator

A1: $x=\dfrac{9\sqrt{5}+15}{2}$ or simplified equivalent

(ii)

M1: Attempts to rearrange the equation to the form $e^{\cdots}=A$ or alternatively takes lns of both sides

M1: Takes lns of both sides and proceeds to find an expression for x, or alternatively applies the laws of logs correctly and proceeds to find an expression for x

A1: $x=\dfrac{2}{7}\left(1+\ln 5\right)$ oe

Question	Scheme	Marks	AOs
6(a)	$(x+4)^2+(y-3)^2$	M1	1.1b
(i)	Centre $(-4,\ 3)$	A1	1.1b
(ii)	Radius is $\sqrt{32}$	A1	1.1b
		(3)	
(b)	$x^2+(2x+k)^2+8x-6(2x+k)-7=0 \Rightarrow ...x^2 \pm ...x \pm ... = 0$	M1	1.1b
	$b^2-4ac=(4k-4)^2-4\times5\times(k^2-6k-7)$	dM1	2.1
	$k^2-22k-39=0 \Rightarrow k=...$	M1	3.1a
	$k=11-4\sqrt{10}$ only	A1	2.2a
		(4)	
			(7 marks)

Notes

(a)

M1: Attempts to complete the square. Score for $(x\pm4)^2...(y\pm3)^2$

A1: $(-4,\ 3)$

A1: $\sqrt{32}$ $\left(\text{or } 4\sqrt{2}\right)$

(b)

M1: Substitutes $y=2x+k$ into the equation of the circle and proceeds to a three-term quadratic (3TQ) in x where the coefficients of "b" and "c" are both in terms of k

dM1: Attempts to find b^2-4ac for a 3TQ where the coefficients of "b" and "c" are both in terms of k. It is dependent on the first method mark.

M1: A complete method to find a value for k. They must have substituted $y=2x+k$ into the equation of the circle, attempted to find the discriminant and attempted to solve the resulting quadratic, in terms of k, set equal to zero.

A1: $k=11-4\sqrt{10}$ or equivalent only. They must have rejected the positive solution.

There may be other geometrical methods to solving (b) e.g.

M1: As the tangent line $y=2x+k$ has gradient 2 , the gradient between the centre of the circle and the point where the line $y=2x+k$ touches C is $-\dfrac{1}{2}$.

$\Rightarrow (2t)^2+t^2=32$ for a parameter t and attempts to solve $(2t)^2+t^2=32$ leading to a value for t $\left(=\dfrac{4\sqrt{10}}{5}\right)$

dM1:	Substitutes their value of t to find the coordinates of the point where the tangent touches the circle. $\left(-4+2"t", 3-"t"\right)$
M1:	A complete method to find a value of k using $k = y - 2x$
A1:	$k = 11 - 4\sqrt{10}$

Question	Scheme	Marks	AOs
7(a)	2^7 or 128 as the constant term	B1	1.1b
	$\left(2-\frac{3x}{4}\right)^7 = ...+{}^7C_1(2)^6\left(-\frac{3x}{4}\right)+{}^7C_2(2)^5\left(-\frac{3x}{4}\right)^2+{}^7C_3(2)^4\left(-\frac{3x}{4}\right)^3+...$ $=...+7\times(2)^6\left(-\frac{3x}{4}\right)+21\times(2)^5\left(-\frac{3x}{4}\right)^2+35\times(2)^4\left(-\frac{3x}{4}\right)^3+...$	M1 A1	1.1b 1.1b
	$=128-336x+378x^2-\frac{945}{4}x^3+...$	A1	1.1b
		(4)	
(b)	Coefficient of x^2 is $"-336"+5\times"378"$	M1	3.1a
	$=1554$	A1	1.1b
		(2)	
			(6 marks)

Notes

(a)

B1: Sight of 2^7 or 128 as the constant term

M1: An attempt at the binomial expansion. This can be awarded for the correct structure of the 2nd, 3rd or 4th term. The correct binomial coefficient must be associated with the correct power of 2 and the correct power of $\pm\frac{3x}{4}$

A1: For a correct simplified or unsimplified second or fourth term (with binomial coefficients evaluated)

A1: $128-336x+378x^2-\frac{945}{4}x^3$ which may be written as a list

(b)

M1: A correct strategy for the required coefficient.

A1: 1554 cao

Question	Scheme	Marks	AOs
8(a)	$(2x-1)^2 = (x-1)^2 + (x+3)^2 - 2(x-1)(x+3)\cos 60°$ oe	M1	3.1a
	Uses $\cos 60° = \frac{1}{2}$, expands the brackets and proceeds to a 3TQ	dM1	1.1b
	$x^2 - 2x - 4 = 0$ *	A1*	2.1
		(3)	
(b)	$(x =)\ 1+\sqrt{5}$	B1	3.2a
	Area $= \dfrac{1}{2} \times \sqrt{5} \times (4+\sqrt{5}) \times \sin 60°$	M1	1.1a
	Area = awrt 6.04 (cm^2)	A1	1.1b
		(3)	
			(6 marks)

Notes

(a)

M1: Recognises the need to apply the cosine rule and attempts to use it with sides in the correct positions and the formula applied correctly.

dM1: Uses $\cos 60° = \frac{1}{2}$, which may be implied, expands the brackets and proceeds to a 3-term quadratic with terms on one side.

A1*: Obtains the correct quadratic equation with no errors seen.

(b)

B1: Deduces that the value of x is $1+\sqrt{5}$. May be implied by the value used in their attempt to find the area of the triangle.

M1: Attempt to find the area of the triangle with the correct lengths used. The expression is sufficient for this mark.

A1: awrt 6.04 (cm^2) Condone lack of units

Question	Scheme	Marks	AOs
9(a)	$\log_{10} N = 2.2 \pm ....$ or $m = \dfrac{1.7-2.2}{40}$	M1	1.1b
	$\log_{10} N = 2.2 - 0.0125t$	A1	1.1b
		(2)	
(b)	$a = 10^{2.2}$	B1	1.1b
	$b = 10^{"-0.0125"}$	M1	3.1b
	$N = 158 \times 0.972^t$	A1	3.3
		(3)	
(c)(i)	The number of fish in thousands when monitoring began	B1	3.4
(ii)	The rate of decrease of fish in the lake from one year to the next	B1	3.4
		(2)	
(d)	119 000	B1	2.2a
		(1)	
(e)	$20 = 158 \times 0.972^T \Rightarrow T = \log_{0.972}\left(\dfrac{20}{158}\right)$ or $\log_{10} 20 = 2.2 - 0.0125T \Rightarrow T = \dfrac{2.2 - \log_{10} 20}{0.0125}$	M1	3.4
	$T =$ awrt 72 (or accept awrt 73)	A1	1.1b
		(2)	
(f)	Suggests the rate of decrease may not remain constant	B1	3.5b
		(1)	
			(11 marks)

Notes

(a)
M1: Attempts to find the gradient of the line or proceeds to a linear equation of the form $\log_{10} N = 2.2 \pm ....$
A1: $\log_{10} N = 2.2 - 0.0125t$

(b)
B1: Correct expression for *a*, may be implied by correct value
M1: Attempts to find *b*
A1: $N = 158 \times 0.972^t$

(c)
(i)
B1: The initial number of fish in thousands when the population was first recorded
(ii)
B1: See scheme, e.g. allow 'The number of fish is decreasing by 2.8% a year'.

(d)
B1: 119 000 cao

(e)
M1: Sets up a correct equation and proceeds to find a value for *T*. This mark cannot be implied by the correct answer. Condone use of 20 000 here.
A1: awrt 72 or awrt 73

(f)

B1: Gives a valid reason suggesting that the rate of decrease may change.

e.g. The temperature could change which would change the rate of decrease.

e.g. The amount of food for the fish may increase which would reduce the rate of decrease.

e.g. The reason for the decrease may change such as not as many are being caught.

Question	Scheme	Marks	AOs
10(a)	$2x^2-7x+8=-3x+14 \Rightarrow 2x^2-4x-6=0$	M1 A1	1.1b 1.1b
	$x=3$	A1	2.2a
		(3)	
(b)	$\int 2x^2-7x+8\ \mathrm{d}x=\frac{2x^3}{3}-\frac{7}{2}x^2+8x\quad(+c)$ or $\int 4x+6-2x^2\ \mathrm{d}x=2x^2+6x-\frac{2}{3}x^3\quad(+c)$	M1 A1	1.1b 1.1b
	Area $=\frac{(14+"5")}{2}\times"3"-\left[\frac{2x^3}{3}-\frac{7}{2}x^2+8x\right]_0^{"3"}=...$ or Area $=\left[2x^2+6x-\frac{2}{3}x^3\right]_0^{"3"}=...$	M1	3.1a
	Area = 18	A1	1.1b
		(4)	
		(7 marks)	

Notes

(a)

M1: Sets the curve equal to the line and rearranges to form a 3TQ

A1: $2x^2-4x-6=0$ oe

A1: $x=3$

(b)

M1: Attempts to integrate the curve or alternatively the line-curve. Award for increasing the power by 1 on one of the terms. Allow slips in collecting like terms in the alternative method.

A1: Correct integrated expression (ignore any reference to +*c*)

M1: The overall strategy to find the shaded area proceeding to find a value for the area. In the method using the area of the trapezium, they must have attempted to find the *y* coordinate of *P*

A1: 18 cao

Question	Scheme	Marks	AOs
11(a)	200 (miles)	B1	3.4
		(1)	
(b)	$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{3}{250}x^2-\frac{6}{5}x+24$	M1 A1	3.4 1.1b
	$\frac{\mathrm{d}y}{\mathrm{d}x}=0\Rightarrow\frac{3}{250}x^2-\frac{6}{5}x+24=0\Rightarrow x=...$	M1	1.1b
	$x=\text{awrt }27.6\text{ only}$	A1	2.3
		(4)	
(c)	$\left(\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=\right)\frac{3}{125}x-\frac{6}{5}=\frac{3}{125}(\text{"}27.6\text{"})-\frac{6}{5}$	M1	1.1b
	$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=(-0.5...)<0\Rightarrow$ Hence y is maximised	A1	2.4
		(2)	
(d)	Maximum distance $=\frac{1}{250}(\text{"}27.6\text{"})^3-\frac{3}{5}(\text{"}27.6\text{"})^2+24(\text{"}27.6\text{"})$	M1	3.4
	Maximum distance $=\text{awrt }289\text{(miles)}$	A1	1.1b
		(2)	
			(9 marks)

Notes

(a)

B1: 200 (miles)

(b)

M1: Attempts to differentiate $y=\frac{1}{250}x^3-\frac{3}{5}x^2+24x$ (decreases the power by one on at least one of their terms)

A1: $\frac{3}{250}x^2-\frac{6}{5}x+24$

M1: Sets their derivative = 0 and attempts to solve their 3TQ to find a value for x

A1: awrt 27.6 only (they must reject 72.4 if found)

(c)

M1: Attempts to differentiate their quadratic to achieve a linear expression and substitutes in their value for x or considers the sign of the second derivative.

A1: Correct derivative, calculation and conclusion

(d)

M1: Substitutes in their value for x into the original equation to find a value for y

A1: awrt 289

Question	Scheme	Marks	AOs
12	$\log_x 81 + \log_6 9 = \log_x 27 - \log_6 4$	M1	1.1b
	$\log_x 3 = -\log_6 36$	M1	1.1b
	$\log_x 3 = -2$	A1	1.1b
	$x^{-2} = 3 \Rightarrow x = ...$	M1	2.1
	$x = \dfrac{\sqrt{3}}{3}$ oe	A1	1.1b
		(5)	
			(5 marks)

Notes

Note: Candidates are told they should not use a calculator for this question, so all stages of working must be seen.

M1: Attempts to use the power rule on at least one of the terms

M1: Attempts to use the addition or subtraction laws of logarithms at least once

A1: Correct equation (may be implied)

M1: Removes the log correctly and finds a value for x

A1: $x = \dfrac{\sqrt{3}}{3}$ or $\dfrac{1}{\sqrt{3}}$ cso

Note: All previous M marks must have been scored, in particular the previous M mark cannot be implied i.e. an index equation, rather than a log equation, must be seen.

Question	Scheme	Marks	AOs
13(a)	e.g. $\frac{\tan\theta\left(16+9\sin^2\theta\right)}{3\sin\theta+5\tan\theta}=\frac{16+9\sin^2\theta}{3\cos\theta+5}$	M1	2.1
	$=\frac{16+9(1-\cos^2\theta)}{3\cos\theta+5}$	M1	1.1b
	$\frac{25-9\cos^2\theta}{3\cos\theta+5}=\frac{(5-3\cos\theta)(5+3\cos\theta)}{3\cos\theta+5}$	M1	1.1b
	$=5-3\cos\theta$ *	A1*	2.2a
		(4)	
(b)	$5-3\cos 2x=\frac{4}{\cos 2x}-3\Rightarrow ...\cos^2 2x+...\cos 2x+...=0$	M1	1.1b
	$3\cos^2 2x-8\cos 2x+4=0$	A1	1.1b
	$\cos 2x=\frac{2}{3}\Rightarrow x=...$	M1	2.1
	$x=\text{awrt }24.1,\quad \text{awrt }155.9$	A1	1.1b
		(4)	
		(8 marks)	

Notes

(a) There are various methods to prove the identity so the method marks may be awarded in a different order to how they have been listed below:

M1: Uses $\tan\theta=\frac{\sin\theta}{\cos\theta}$ to achieve an expression in $\sin\theta$ and $\cos\theta$

M1: Attempts to use $\pm\sin^2\theta\pm\cos^2\theta=\pm 1$ to achieve an expression in $\cos\theta$ only

M1: Factorises $\left(9\cos^2\theta-25\right)$ into $\left(3\cos\theta+5\right)\left(3\cos\theta-5\right)$ or alternatively they may attempt to multiply $\left(3\sin\theta+5\tan\theta\right)\left(5-3\cos\theta\right)$ leading to four terms.

A1*: Achieves the given answer with all intermediate stages shown or alternatively if they multiply both sides by $3\sin\theta+5\tan\theta$ then look for both sides being equal and either a conclusion or a preamble with minimal conclusion e.g. tick/QED

(b)

M1: Sets $5-3\cos 2x=\frac{4}{\cos 2x}-3$ and rearranges the equation to achieve a 3-term quadratic in $\cos 2x$

A1: $3\cos^2 2x-8\cos 2x+4=0$

M1: A correct method to solve their 3-term quadratic in $\cos 2x$ and proceeds to find a value for x

A1: $x=\text{awrt }24.1,\quad \text{awrt }155.9$ and no others in the range

Question	Scheme	Marks	AOs
14(a)	$\dfrac{25x^2-30x-9}{2x^{\frac{1}{2}}}=...x^{\frac{3}{2}}\pm...x^{\frac{1}{2}}\pm...x^{-\frac{1}{2}}$	M1	1.1b
	$\dfrac{25}{2}x^{\frac{3}{2}}-15x^{\frac{1}{2}}-\dfrac{9}{2}x^{-\frac{1}{2}}$	A1	1.1b
	$\displaystyle\int\frac{25}{2}x^{\frac{3}{2}}-15x^{\frac{1}{2}}-\frac{9}{2}x^{-\frac{1}{2}}\,\mathrm{d}x=...x^{\frac{5}{2}}\pm...x^{\frac{3}{2}}\pm...x^{\frac{1}{2}}\quad(+c)$	M1	1.1b
	$5x^{\frac{5}{2}}-10x^{\frac{3}{2}}-9x^{\frac{1}{2}}+c$	A1	1.1b
		(4)	
(b)(i)	$5k^{\frac{5}{2}}-10k^{\frac{3}{2}}-9k^{\frac{1}{2}}-(-14)<6\sqrt{k}+14$	M1	1.1b
	$5k^{\frac{5}{2}}-10k^{\frac{3}{2}}-15k^{\frac{1}{2}}<0\Rightarrow k^{\frac{1}{2}}(.......)<0$	dM1	2.1
	$k^2-2k-3<0\quad *$	A1*	1.1b
(ii)	Critical values $-1,\ 3\Rightarrow k<3$	M1	1.1b
	$2<k<3$	A1	3.2a
		(5)	
			(9 marks)

Notes

(a)

M1: Expands the bracket and attempts to split the fraction up into separate terms and achieves at least one correct index on one term from correct work

A1: Two correct of $\dfrac{25}{2}x^{\frac{3}{2}}, -15x^{\frac{1}{2}}, -\dfrac{9}{2}x^{-\frac{1}{2}}$ which may be simplified or unsimplified

M1: Increases the power of x by one on at least one term (which must be a fractional power)

A1: $5x^{\frac{5}{2}}-10x^{\frac{3}{2}}-9x^{\frac{1}{2}}+c$ (including the $+c$)

(b)(i)

M1: Substitutes in the limits k and 1, subtracts either way round and sets less than $6\sqrt{k}+14$. Do not be concerned if an equals or an inequality sign is used.

dM1: Attempts to rearrange their inequality and either takes out a factor or cancels by $k^{\frac{1}{2}}$ leading to a 3TQ. Do not be concerned if an equals or an inequality sign is used. It is dependent on the first method mark.

A1*: $k^2-2k-3<0$ with no errors seen and all stages of working shown

(ii)

M1: Finds the critical values for the given inequality and selects $k<a$ where a is their larger critical value. Condone e.g. $-1<k<a$

A1: $2<k<3$ only (or equivalent)

Question	Scheme	Marks	AOs
15	$(x+7)(x+1)>2x-3\Rightarrow x^2+8x+7>2x-3\Rightarrow x^2+6x+10>0$	M1 A1	2.1 1.1b
	$(x+3)^2+1>0$ or $b^2-4ac=6^2-4\times1\times10<0$	M1	2.1
	e.g. $(x+3)^2+1>0$ or $b^2-4ac=-36$ with full reasoning Hence $(x+7)(x+1)>2x-3$ (for all $x\in\mathbb{R}$) *	A1*	2.4
		(4 marks)	

Notes

M1: Attempts to multiply out and collect terms to achieve a 3TQ:
$(x+7)(x+1)>2x-3\Rightarrow x^2+8x+7>2x-3\Rightarrow ...x^2+...x+...>0$ oe

A1: $x^2+6x+10>0$ oe

M1: Shows that $(x+3)^2+1>0$ by completing the square or shows that the discriminant of their quadratic is negative

A1*: Either

- Explains that as $(x+3)^2\geq 0$ for all $x\in\mathbb{R}$ then $(x+3)^2+1>0$ so $(x+7)(x+1)>2x-3$
- Explains that since $x^2+6x+10$ is a positive quadratic and as the discriminant is negative (-36) there are no real roots hence $(x+7)(x+1)>2x-3$ (for all $x\in\mathbb{R}$)

Alternative proof

M1: Starts the proof with $(x+3)^2\geq 0$ and attempts to multiply out to achieve a 3-term quadratic

A1: $x^2+6x+9\geq 0$

M1: Shows that $x^2+8x+7\geq 2x-2$

A1*: Explains that $x^2+8x+7>2x-3$ and hence $(x+7)(x+1)>2x-3$ (for all $x\in\mathbb{R}$)

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